

# FineFlow–Espresso

A reduced one-dimensional model for fines release, migration, blockage,  
detachment and cup washout

Technical model and numerical documentation

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The parameter values shown here are illustrative and uncalibrated.

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## 1 Purpose and scope

FineFlow–Espresso is a transparent continuum demonstrator for the coupled hydraulics and transport of coffee fines through an espresso puck. The main objective is to predict how initially attached fines are released, transported, deposited, and possibly detached again; how these processes alter local permeability; and how the changing resistance affects machine pressure and flow. The model additionally distinguishes fines leaving the puck from fines retained by the portafilter basket and fines entering the cup.

The model is deliberately one-dimensional. It is intended to organize measurements, test mechanistic hypotheses, and provide interpretable predictions without resolving each pore. It is not a substitute for pore-scale simulation. Computed axial blockage locations are effective model states rather than direct predictions of lateral channel geometry. **Reverse flow and radial channeling are outside the scope of this work.** Local permeability recovery describes deposit removal, not radial bypass flow.

The software supports three levels of structural information:

1. uniform initial porosity and permeability;
2. depth-resolved initial profiles obtained from measurements or image analysis;
3. a tabulated deposit–permeability relationship supplied by pore-network modelling.

## 2 Physical picture and system boundary

Water enters at the top of a puck of length  $L$  and cross-sectional area  $A$ . The axial coordinate  $z$  points from the inlet,  $z = 0$ , to the basket,  $z = L$ . Fines can move among three puck inventories: available fines attached to the coffee matrix, mobile fines in the pore liquid, and deposited fines. At the puck outlet the mobile-fines flux is divided between a basket-retained inventory and a cup inventory.

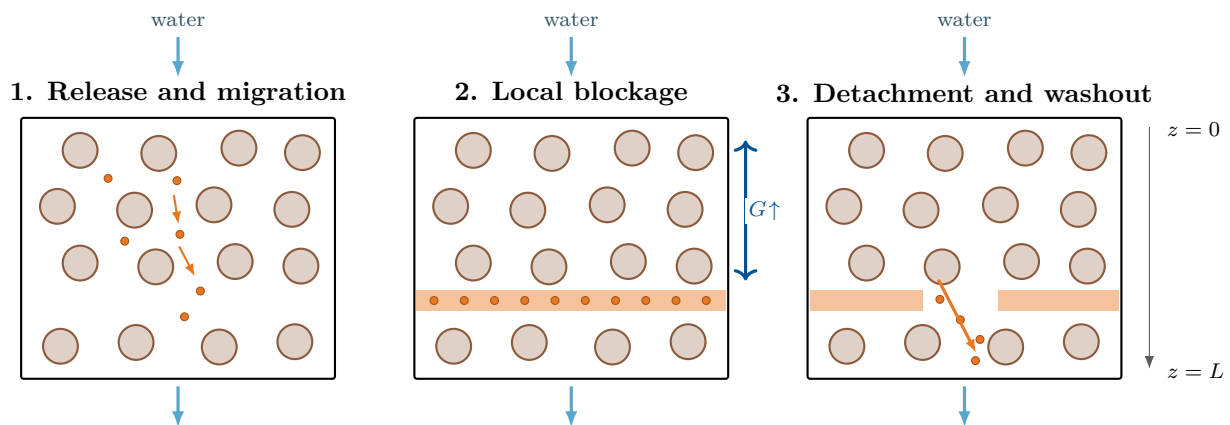


Figure 1: Effective one-dimensional mechanism. Mobile fines occupy open voids, while an unresolved deposited layer is represented by a local mass inventory and reduced effective permeability. If hydraulic loading exceeds an effective strength, deposits detach. Only the fraction that can escape through the local pore passages is returned to the mobile phase.

### 2.1 Principal assumptions

- Water and suspended fines are incompressible, and the puck area is constant.
- The superficial velocity  $q(t) = Q(t)/A$  is spatially uniform; local pore velocity is  $q/\epsilon$ .
- Radial flow redistribution and explicit channels are not resolved.

- One effective fines population is used; particle-size-dependent transport is deferred.
- Fluid properties are constant and extraction chemistry, gas release, swelling, and thermal variation are omitted.
- Darcy flow (default), with optional Forchheimer inertia, is evaluated as an algebraic pressure-gradient closure; no momentum PDE is solved.
- The basket model is level 1: constant hydraulic resistance and constant fines-retention fraction.
- Basket deposits do not yet modify basket resistance and cannot detach.
- Empirical closure parameters are effective quantities and require calibration.

### 3 State variables and conservation laws

The puck stores three fines inventories per unit bulk puck volume:

| Symbol                | Unit               | Meaning                                                |
|-----------------------|--------------------|--------------------------------------------------------|
| $m_a(z, t)$           | $\text{kg m}^{-3}$ | available fines still attached to the coffee matrix    |
| $\varepsilon c(z, t)$ | $\text{kg m}^{-3}$ | mobile fines per bulk volume; $c$ is per liquid volume |
| $m_d(z, t)$           | $\text{kg m}^{-3}$ | deposited fines retained in the pore space             |

The continuum balances are

$$\frac{\partial(\varepsilon c)}{\partial t} + \frac{\partial(qc)}{\partial z} = \frac{\partial}{\partial z} \left( \varepsilon D \frac{\partial c}{\partial z} \right) + R_{\text{rel}} - R_{\text{dep}} + R_{\text{wash}}, \quad (1)$$

$$\frac{\partial m_d}{\partial t} = R_{\text{dep}} - R_{\text{wash}}, \quad (2)$$

$$\frac{\partial m_a}{\partial t} = -R_{\text{rel}}. \quad (3)$$

Here  $D$  is an effective axial dispersion coefficient. Integrating the three equations over the puck shows that reaction terms cancel. Consequently, the initial fines mass equals the sum of available, mobile, deposited, basket-retained, and cup fines:

$$M_0 = M_a + M_m + M_d + M_{\text{sieve}} + M_{\text{cup}}. \quad (4)$$

## 4 Constitutive and kinetic models

### 4.1 Release

The available fines decay with a flow-dependent first-order rate:

$$R_{\text{rel}} = k_{\text{rel}} m_a f_q(q), \quad (5)$$

$$f_q(q) = \left( \frac{|q|}{q_{\text{ref}}} \right)^{a_r}. \quad (6)$$

The exact exponential fraction released over a numerical step  $\Delta t$  is  $1 - \exp[-k_{\text{rel}} f_q \Delta t]$ . This update is nonnegative and avoids an additional reaction-stability restriction.

## 4.2 Deposition

The deposition closure is

$$R_{\text{dep}} = k_{\text{dep}} \chi(z) |q| c \left[ 1 - \frac{m_{\text{d}}}{m_{\text{cap}}} \right]_+, \quad (7)$$

where  $k_{\text{dep}}$  is a filtration coefficient,  $m_{\text{cap}}$  is the local deposit capacity, and

$$\chi(z) = 1 + (\chi_{\text{out}} - 1) \exp \left[ -\frac{L - z}{\ell_{\text{out}}} \right] \quad (8)$$

is an optional empirical increase in capture near the basket. The latter is a one-dimensional representation of stronger retention near the outlet and must not be interpreted as resolved basket-hole geometry.

## 4.3 Hydraulic detachment and geometric escape

The revised model separates adhesion failure from successful remobilization. Hydraulic loading first produces a potential detachment rate,

$$R_{\text{det}}^* = k_{\text{det}} m_{\text{d}} \left[ \frac{G}{G_{\text{crit}}} - 1 \right]_+, \quad (9)$$

$$G = \left| \frac{\partial p}{\partial z} \right|. \quad (10)$$

The local hydraulic diameter is either supplied by CT/PNM or initialized from the packed-bed surface relation

$$d_{\text{h},0} = \frac{2}{3} \phi_s d_{\text{eff}} \frac{\varepsilon_0}{1 - \varepsilon_0}. \quad (11)$$

With representative fine diameter  $d_f$ , the confinement ratio is  $\lambda = d_f/d_{\text{h}}$ . The net transfer to the mobile phase is

$$R_{\text{wash}} = P_{\text{esc}}(\lambda) R_{\text{det}}^*. \quad (12)$$

The default smooth closure is

$$P_{\text{esc}}(\lambda) = \frac{1}{1 + \exp[s(\lambda - \lambda_c)]}, \quad (13)$$

where  $\lambda_c$  is the 50% escape ratio and  $s$  controls transition sharpness. Alternatively, the code accepts a PNM table of size ratio versus escape probability. Potentially detached fines that do not escape are treated as immediately recaptured locally. This avoids the discontinuity and weak identifiability of a hard  $d_f/d_h$  cutoff.

## 4.4 Porosity and permeability

Deposited fines reduce liquid-filled porosity according to

$$\varepsilon(z, t) = \max \left[ \varepsilon_0(z) - \frac{m_{\text{d}}}{\rho_{\text{d}}}, \varepsilon_{\text{min}} \right]. \quad (14)$$

The software provides three alternative permeability closures.

**Exponential closure.**

$$\frac{K}{K_0} = \max \left[ \exp \left( -\beta \frac{m_{\text{d}}}{m_{\text{cap}}} \right), \frac{K_{\text{min}}}{K_0} \right]. \quad (15)$$

The exponential is an empirical demonstrator choice, not a coffee-specific law derived from measurements.

Writing  $x = m_d/m_{\text{cap}}$ , a constant logarithmic sensitivity  $d \ln(K/K_0)/dx = -\beta$  integrates to Eq. (15) with  $K/K_0 = 1$  at  $x = 0$ . This motivates a smooth, positive and tunable closure; it does not establish its physical validity. At capacity, the illustrative  $\beta = 5.5$  imposes  $K/K_0 = e^{-5.5} = 0.00409$  before clipping. Strong blockage is thus an input assumption.

### Kozeny–Carman-style closure.

$$\frac{K}{K_0} = \left( \frac{\varepsilon}{\varepsilon_0} \right)^3 \left( \frac{1 - \varepsilon_0}{1 - \varepsilon} \right)^2, \quad (16)$$

subject to clipping between the configured minimum ratio and one. This is a relative porosity law with independently prescribed  $K_0(z)$ . It assumes the surface/geometry prefactor is unchanged during deposition. A constant Kozeny factor cancels when dividing by the initial state; the implementation has no separate Kozeny-factor input. The exponential parameter `permeability_blocking_factor` is not a Kozeny factor and is unused here. Deposit density determines porosity loss, while deposit capacity still limits capture. Preferential throat blockage and changing specific surface are not resolved by this relative law.

**Pore-network table.** For `permeability_model = pnm_table`, the code linearly interpolates a user-provided table of  $m_d$  versus  $K/K_0$ . This allows a pore-network calculation to define the continuum closure without forcing it into the exponential form. Values outside the supplied deposit range use the nearest table endpoint and are then clipped between the minimum permeability ratio and one.

## 5 Hydraulics, machine control, and basket sieve

### 5.1 Darcy baseline and optional Forchheimer resistance

No momentum equation is solved. For constant area and incompressible flow, the superficial velocity  $q = Q/A$  is uniform in  $z$ , and the local pressure-gradient magnitude is evaluated algebraically:

$$G(z, t) = \frac{\mu q}{K(z, t)} + \rho \beta_F(z, t) |q| q. \quad (17)$$

The first term is viscous Darcy resistance. The second is nonlinear inertial resistance, with Forchheimer coefficient  $\beta_F$  in  $\text{m}^{-1}$ . The default `hydraulic_model = darcy` sets  $\beta_F = 0$ . The optional evolution law

$$\beta_F = \beta_{F,0} \left( \frac{K_0}{K} \right)^{n_F} \quad (18)$$

is available, but  $n_F = 0$  is recommended until PNM or experiments demonstrate systematic evolution during blockage.

For positive flow, axial integration gives

$$\Delta p_{\text{puck}} = q \int_0^L \frac{\mu}{K} dz + q^2 \int_0^L \rho \beta_F dz. \quad (19)$$

Consequently, prescribed-pressure operation requires only the positive root of a scalar quadratic equation when inertia is enabled, or a linear resistance inversion for Darcy flow.

### 5.2 Measured permeability and the role of Ergun

Ergun estimates the complete resistance from porosity, sphericity and one effective particle diameter. CT and the five-parameter PSD can supply these quantities, but they do not retain connectivity,

tortuosity, bottlenecks or localized throat blockage. In Eq. (17),  $K(z, t)$  can instead be taken directly from CT-based PNM or low-flow experiments, while  $\beta_F$  represents only the additional nonlinear resistance. The linear and quadratic parts can be separated experimentally from  $\Delta p = c_1 Q + c_2 Q^2$  using several imposed flows. Ergun remains useful as a baseline and gives the estimate

$$\beta_{F,\text{Ergun}} = \frac{1.75(1 - \varepsilon)}{\varepsilon^3 d_{\text{eff}}}. \quad (20)$$

Darcy is selected for the current tested regime. The optional inertial term should be enabled when pressure–flow data require it. The comparison below is not evidence that inertia is negligible for every grind or flow rate.

### 5.3 Level-1 basket boundary

The clean basket is a constant series resistance  $R_s$ :

$$\Delta p_{\text{puck}} = \int_0^L G \, dz, \quad (21)$$

$$\Delta p_{\text{sieve}} = Q R_s, \quad (22)$$

$$\Delta p_{\text{total}} = \Delta p_{\text{puck}} + \Delta p_{\text{sieve}}. \quad (23)$$

If a fraction  $\eta_s$  of puck-outlet fines is retained, then

$$\dot{M}_{\text{cup}} = (1 - \eta_s) \dot{M}_{\text{out,puck}}, \quad (24)$$

$$\dot{M}_{\text{sieve}} = \eta_s \dot{M}_{\text{out,puck}}. \quad (25)$$

No sieve detachment term is present at level 1. Therefore, retention changes the reported destination of fines but does not feed back into  $R_s$ .

### 5.4 Hydraulic operating modes

**Flow mode**  $Q(t)$  is prescribed and Eq. (23) predicts total pressure.

**Pressure mode**

Total pressure is prescribed and the positive root of the selected resistance law gives  $Q$  (linear for Darcy, quadratic with inertia). An optional pressure pulse can challenge deposit stability.

**PI mode** The machine changes its flow command to approach a total-pressure target, subject to a pump-flow cap and actuator lag.

The PI implementation combines resistance-based feedforward with proportional and integral correction. With pressure error  $e = (p_{\text{set}} - p)/10^5$  in bar,

$$\Delta p(Q_{\text{ff}}) = p_{\text{set}}, \quad (26)$$

$$Q_{\text{raw}} = Q_{\text{ff}} + K_P e + K_I \int e \, dt, \quad (27)$$

$$Q_{\text{cmd}} = \min[\max(Q_{\text{raw}}, 0), Q_{\text{max}}]. \quad (28)$$

The actual flow approaches the command with first-order time constant  $\tau_p$ . Conditional integration suppresses windup while the command is saturated. A linear startup ramp can be applied to  $p_{\text{set}}$ . Thus a sufficiently permeable puck may stay below 9 bar when  $Q_{\text{max}}$  is reached.

## 6 Initial and boundary conditions

### 6.1 Initial conditions

Unless axial arrays are supplied, all cells begin with uniform  $\varepsilon_0$ ,  $K_0$ ,  $m_{a,0}$ ,  $m_{d,0}$  and  $c_0$ . Optional arrays `initial_porosity_profile` and `initial_permeability_profile_m2` must each contain exactly the number of finite-volume cells. These arrays can represent depth-averaged results of CT segmentation and pore-network flow calculations.

### 6.2 Fines-transport boundaries

At the water inlet, the incoming water contains no fines:

$$(qc)_{z=0} = 0. \quad (29)$$

The code imposes zero dispersive flux at both ends,

$$\varepsilon D \frac{\partial c}{\partial z} \Big|_{z=0,L} = 0, \quad (30)$$

and uses a convective outflow at  $z = L$ :

$$\dot{M}_{\text{out,puck}} = Aqc(L, t). \quad (31)$$

Reverse flow is not supported by the present demonstrator.

### 6.3 Pressure boundaries

The downstream cup is the zero-gauge-pressure reference. The pressure just above the sieve is  $\Delta p_{\text{sieve}}$ . The reconstructed puck pressure therefore decreases from total inlet pressure to sieve pressure rather than to zero. The remaining drop occurs across the lumped sieve resistance.

## 7 Finite-volume discretization

The interval  $[0, L]$  is divided into  $N$  control volumes of width  $\Delta z = L/N$ . Cell-centred states are indexed by  $i = 1, \dots, N$ , while advective and dispersive fluxes are evaluated at faces  $i \pm \frac{1}{2}$ .

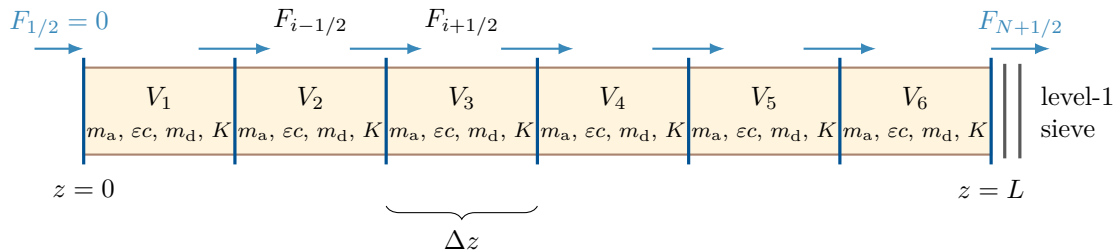


Figure 2: One-dimensional finite-volume mesh. Reaction states are cell centred; mobile-fines fluxes are face centred. The sieve is a lumped boundary element after the last puck cell.

Writing  $u_i = (\varepsilon c)_i$ , the mobile balance is discretized as

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = -\frac{F_{i+1/2} - F_{i-1/2}}{\Delta z} + R_{\text{rel},i} - R_{\text{dep},i} + R_{\text{wash},i}. \quad (32)$$

For forward flow, the face flux is

$$F_{i+1/2} = qc_i - \varepsilon_{i+1/2} D \frac{c_{i+1} - c_i}{\Delta z}. \quad (33)$$

Advection is first-order upwind and dispersion is centred. Each internal face has one flux used with opposite signs in its adjacent cells, so internal fluxes cancel when summing the balances. Equal-and-opposite inventory transfers and consistent outlet/basket accounting preserve total fine mass. This is why FVM is a natural conservative choice; conservation is not guaranteed merely by calling a scheme finite volume. It also does not guarantee accuracy or physical validity. The implementation monitors the global mass residual; the upwind part is monotone under the applied CFL restriction and therefore does not create spurious oscillatory extrema in the pure-advection test.

The local Darcy–Forchheimer gradient is calculated in each cell. The discrete puck pressure drop is

$$\Delta p_{\text{puck}} = \sum_{i=1}^N \left( \frac{\mu q}{K_i} + \rho \beta_{F,i} |q| q \right) \Delta z. \quad (34)$$

Face pressures are reconstructed by cumulative integration of the cell gradients, starting from total inlet pressure and ending at sieve pressure.

## 8 Time integration and solution algorithm

The implementation uses explicit sequential operator splitting during each time step:

1. calculate porosity and permeability from the current deposit field;
2. calculate Darcy–Forchheimer coefficients, flow, total pressure, local gradients and cell pressure;
3. choose a stable time step;
4. update release, deposition, potential detachment and size-selective escape with bounded exponential fractions;
5. update mobile fines conservatively using face fluxes;
6. split puck-outlet fines between basket and cup;
7. update the PI controller and pump actuator when PI mode is active;
8. record states at the requested output time.

The time step is bounded by the user maximum, remaining time to the next output, controller resolution, and a combined advection–dispersion estimate:

$$\Delta t \leq \frac{C_{\text{CFL}}}{|q|/(\varepsilon_{\min} \Delta z) + 2D/\Delta z^2}. \quad (35)$$

In PI mode, an additional limit  $\Delta t \leq 0.25\tau_p$  resolves the actuator. Output times and pressure-pulse switching times are hit exactly.

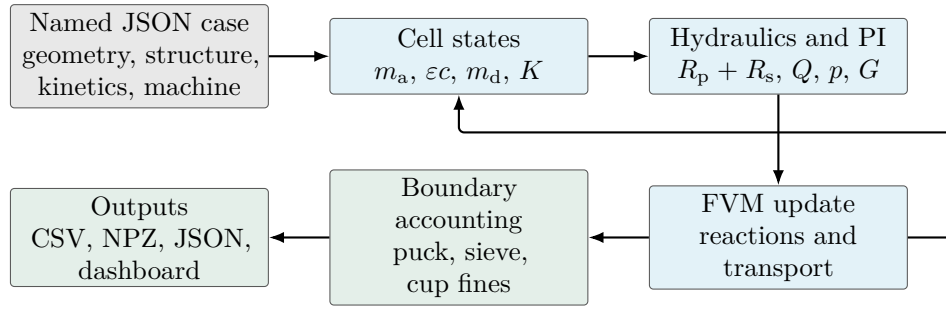


Figure 3: Computational workflow for one named case.

## 9 Configuration and calibration strategy

Each named JSON case is applied on top of the Python defaults. A selected case does not inherit values from another JSON case. Command-line options override both defaults and the selected case. Generated parameter JSON files contain the complete parameter set actually used.

For reproducible calibration, parameters are divided by the information source that can actually identify them.

| Category                 | Parameters or fields                                                                   | Determination method                                                                                                                                                                                                                                                         |
|--------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Measured and fixed       | $L, A, \mu, \rho, \text{dose, PSD, } d_f, d_{\text{eff}}$                              | Direct geometry and temperature measurements. Fit the Bora–Briesen five-parameter bimodal volume PSD; calculate the fraction below a selected mobility cutoff and conditional fine/coarse diameters. The PSD parameters are inputs, not kinetic fit parameters.              |
| CT-derived structure     | $\varepsilon_0(z), d_{h,0}(z), \text{deposit location}$                                | Segment connected void and solid in axial slabs. Calculate $d_h = 4V_{\text{void}}/A_{\text{wetted}}$ and quantify voxel and segmentation uncertainty. Use replicated interrupted shots to observe structural changes.                                                       |
| PNM-derived closures     | $K_0(z), K/K_0(m_d), d_h/d_{h,0}(m_d), D, \text{capture and } P_{\text{esc}}(d_f/d_h)$ | Solve network flow and particle transport on CT-derived networks. Sample the measured PSD, record deposition, breakthrough, conductance loss and escape, then export lookup tables. Scale the global PNM resistance to an independent bulk low-flow measurement if required. |
| Hydraulic experiments    | global $K$ scale, $\beta_F(z), R_s$                                                    | Measure $\Delta p(Q)$ at several imposed flows and fit $\Delta p = c_1 Q + c_2 Q^2$ . Low-flow data constrain $K$ ; a broad flow range is needed for $\beta_F$ . Determine clean-basket resistance separately.                                                               |
| Machine characterization | $Q_{\text{max}}, \tau_p, \text{pressure target, PI gains}$                             | Measure pressure and flow without coffee and with known resistances. Fix these before fitting puck kinetics.                                                                                                                                                                 |

| Category                  | Parameters or fields                                                                     | Determination method                                                                                                                                                                                                |
|---------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dynamic fines experiments | releasable inventory, $k_{\text{rel}}$ , $k_{\text{det}}$ , $G_{\text{crit}}$ , $\eta_s$ | Use time-resolved cup-fines mass or turbidity, preferably outlet PSD, together with pressure and flow. Pressure pulses help separate threshold and rate. Measure basket-retained mass independently where possible. |

The recommended sequence is: (1) fix geometry, fluid properties, dose and PSD; (2) insert CT porosity and hydraulic-diameter profiles; (3) determine  $K$  and  $\beta_F$  from pressure–flow experiments; (4) import PNM blockage, dispersion and escape tables; (5) characterize basket and machine; (6) fit release to early fines washout; and (7) fit detachment to controlled pressure events. Validation must use withheld pucks or grind settings.

A single static CT image cannot determine release or detachment kinetics. Nor can pressure and flow alone separate release, deposition, permeability loss, detachment and basket capture. Fitting all parameters simultaneously would therefore be structurally non-identifiable.

## 10 Controlled comparison and demonstration results

The frozen `comparison_reference.json` defines a 40-cell, 60-s study. From the solver directory, reproduce its three cases with:

```
python3 compare_closures.py
```

The script does not read the active named case configuration. All kinetics, initial permeability, controller and numerical settings are held fixed; no parameters are fitted. Only the indicated switch changes:

1. Forchheimer + exponential: archived reference configuration.
2. Darcy + exponential: remove inertia only, relative to case 1.
3. Darcy + Kozeny–Carman: change permeability law only, relative to case 2.

The script checks these parameter differences, global mass error and mobile positivity. The active `fine_puck` case and Python hydraulic default now use Darcy. Exponential remains the permeability default. The explicit `coarse_puck` configuration retains its earlier hydraulic selection.

### 10.1 Removing inertia has negligible effect in this test

The maximum flow difference between cases 1 and 2 is  $2.9642 \times 10^{-7}$  mL/s, or  $2.9102 \times 10^{-7}$  of the reference peak flow. The maximum total-pressure difference is 0.02034 Pa. These results support using Darcy for this illustrative regime. They do not establish a universal absence of inertial losses in espresso pucks.

## 10.2 Permeability choice strongly changes the prediction

Table 2: Controlled Darcy comparison at 60 s, with no parameter refitting.

| Quantity                   | Exponential | Kozeny–Carman |
|----------------------------|-------------|---------------|
| Final total pressure [bar] | 8.99061     | 9.00031       |
| Final flow [mL/s]          | 0.04535     | 0.89243       |
| Cumulative cup fines [g]   | 0.03993     | 0.34664       |
| Minimum local $K/K_0$      | 0.01263     | 0.72264       |

The PI controller approaches the same pressure target in both cases, but their flow and fines predictions differ substantially. The exponential law imposes strong permeability loss; the porosity-only relative law predicts a milder loss for this deposit density. This is sensitivity evidence, not validation of either closure. Do not tune Kozeny–Carman to reproduce the exponential result as though that result were experimental truth. CT, PNM and independent pressure–flow and fines measurements must constrain the closure.

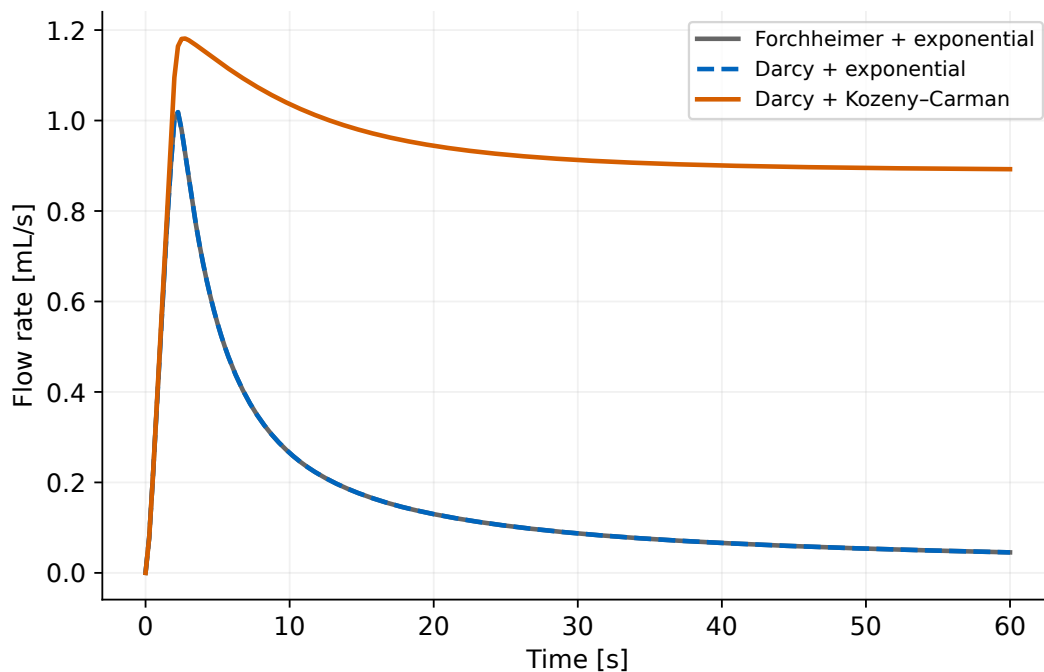


Figure 4: Common-input comparison of the predicted flow histories.

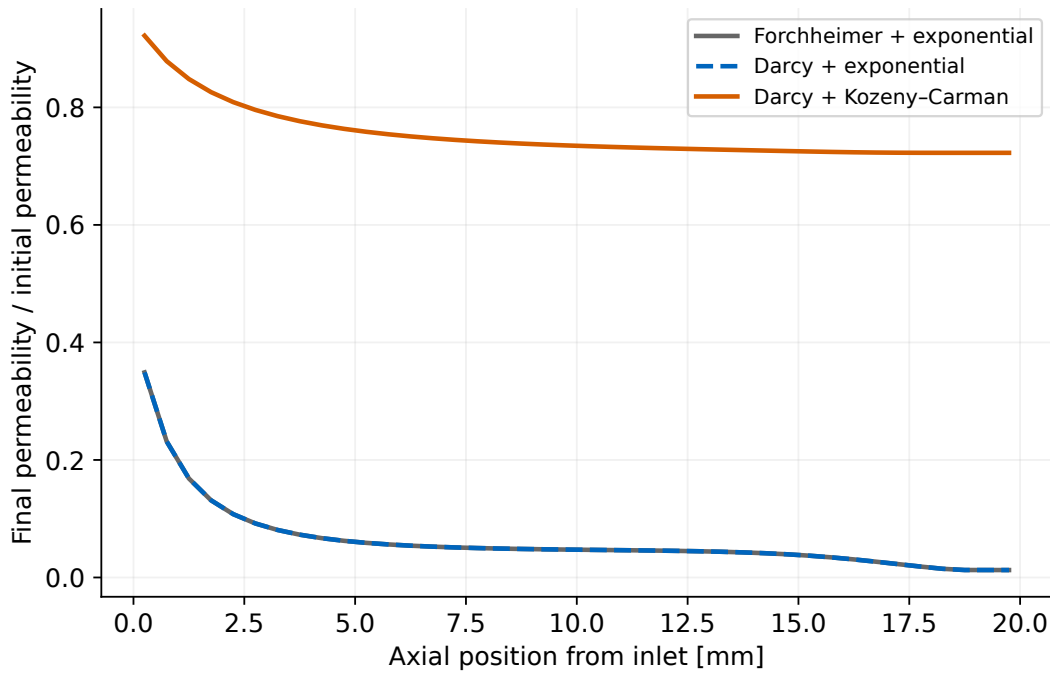


Figure 5: Final relative permeability: closure choice dominates the inertia switch in this test. Initial deposits are zero, so initial permeability equals  $K_0$ .

### 10.3 Comparison metrics and archived outputs

Metrics in `comparison_metrics.json` compare matching saved times and, for fields, all axial cells. For a reference array  $a$  and candidate  $b$ ,

$$E_{\text{abs}} = \max |a - b|, \quad E_{\text{peak}} = \frac{\max |a - b|}{\max |a|}.$$

The normalized value is a fraction, not a percentage. If the reference peak is zero, the script records a null normalized value. Peak normalization avoids pointwise division by zero at startup. These metrics are neither RMSE nor experimental errors. Parameter snapshots, CSV histories, NPZ profiles and single-axis PDF/PNG comparison figures are committed with the study. The older `results/` directory is an archived run, identified by its own parameter snapshot; it should not be described as a fresh run of the new defaults.

## 11 Numerical verification

Verification asks whether the equations are solved correctly; validation asks whether those equations and calibrated parameters represent real espresso. The repository contains conventional unit tests and an extended verification program. The archived verification evidence reports 15 unit tests and 17 extended checks. These checks are numerical verification, not physical validation; re-run them after solver changes.

The extended suite includes:

- comparison with exact uniform-bed Darcy and Darcy–Forchheimer pressure;
- exact nonlinear inversion under prescribed pressure;
- superficial-velocity continuity and finite pore velocity;
- global fines conservation, positivity, porosity and permeability bounds;

- comparison of the release-only model with its analytical exponential solution;
- suppression of successful washout as  $d_f/d_h$  increases;
- total-variation-diminishing and positivity checks for pure advection;
- spatial and temporal refinement studies;
- attainable PI target, pump-cap enforcement, and coarse-bed saturation behaviour.

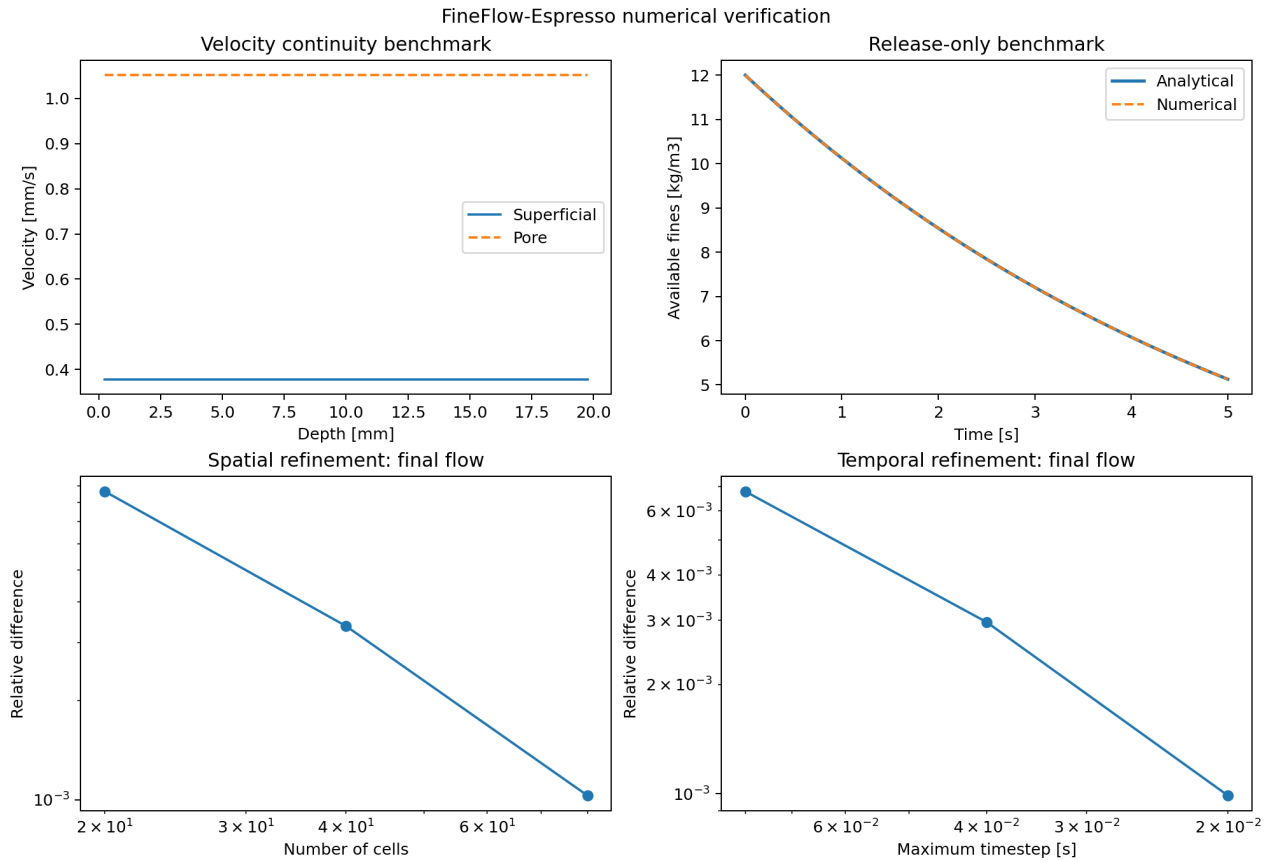


Figure 6: Automatically generated analytical and refinement diagnostics.

At the current settings, the maximum penultimate-to-finest difference across the monitored quantities is approximately 2.9% for spatial refinement and 0.45% for temporal refinement. These are useful software checks, but they do not establish model-form accuracy. Future verification should add a manufactured solution for the complete coupled PDE system and formal observed-order estimates.

## 12 Outputs and reproducibility

For case name `fine_puck`, the program writes:

- `fine_puck_timeseries.csv`: flow, controller command, pressure split, fines rates and cumulative destinations, blockage metrics, and mass error;
- `fine_puck_profiles.npz`: time-dependent axial states, pressures, superficial and pore velocity, pressure gradient, Forchheimer coefficient, hydraulic diameter, size ratio, and escape probability;
- `fine_puck_summary.json`: compact performance and conservation metrics;
- `fine_puck_parameters.json`: complete parameters actually used;
- `fine_puck_dashboard.png`: the four-panel result figure;

- four standalone manuscript panels for hydraulic response, fines washout, deposited fines and relative permeability, each as 600-dpi PNG and vector PDF.

The verification program writes a text report, pass/fail CSV, spatial and temporal convergence tables, summary JSON, and verification figure. A nonzero program exit status indicates that at least one verification criterion failed.

### 13 Known limitations and planned extensions

- One fines class cannot represent size-dependent pore capture and basket passage.
- Level-1 basket resistance does not increase with retained fines.
- Basket-retained fines cannot detach or enter the cup later.
- Deposit-capacity and outlet-capture profiles are empirical.
- Static CT constrains structure but not unique kinetic parameters.
- Reverse flow and radial channeling are outside scope; cracks and individual basket holes are unresolved.
- Extraction, dissolution, temperature, gas, swelling and mechanical compaction are absent.
- The simple constant pump-flow cap should ultimately be replaced by a measured pump curve.
- Uncertainty propagation and parameter-identifiability analysis are not yet implemented.

The recommended next physical extension is a small number of fines-size classes with size-dependent capture and sieve retention. The recommended next inference extension is a calibration driver that reads measured pressure, flow, cup-fines, CT-derived profiles, and PNM tables while marking parameters as fixed, informed by priors, or fitted.

### 14 How to compile and update this report

From the `documentation` directory, compile with

```
latexmk -pdf fineflow_espresso_technical_report.tex
```

The result and verification figures are referenced from the solver output folders. Re-run the Python cases before recompiling when updated figures or values are required. Numerical values in Table 2 are currently entered explicitly and should be updated when the reference case changes.